

Increasing movement during office work at sit-stand workstations: A novel seating device to facilitate transitions

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ARTICLE INFO

Keywords:

Low back pain
Chair design
Sit-stand workstation

ABSTRACT

A novel active office chair (Movably Pro) was designed to facilitate frequent sit-stand movement 1) through auditory and tactile prompts and 2) with minimal-to-no work surface adjustment when transitioning. The purpose of this study was to compare lumbopelvic kinematics, discomfort, and task performance between the novel chair and traditional sitting/standing. Sixteen participants completed three separate 2-h sedentary exposures. Although participants transitioned every 3 min between sitting and standing with the novel chair, productivity was not affected. When standing in the novel chair, the lumbopelvic angles fell in between traditional sitting and standing ($p < 0.01$). Movement and/or postural changes that occurred with the novel chair reduced low back and leg discomfort for pain developers (PDs) ($p < 0.01$). All participants classified as PDs in traditional standing were non-PDs with the novel chair. This intervention was effective in reducing sedentary time without the time loss associated with desk movement.

1. Introduction

Over the past five decades, there has been a steady increase in the prevalence of occupations where approximately 60–80% of work hours are spent in sedentary behaviours (Church et al., 2011; Mork and Westgaard, 2009; Mörl and Bradl, 2013; Parry and Straker, 2013; Ryan et al., 2011; Thorp et al., 2012; Toomingas et al., 2012). Exposure to prolonged sedentary postures (sitting and standing) has been linked to approximately 40%–70% of individuals developing acute low back pain (LBP) (Gallagher et al., 2011; Marshall et al., 2011). In addition, prolonged sedentary work, both sitting and standing, has been linked to several negative health conditions, including lower extremity venous pooling (Smith et al., 2018), atherosclerosis (Krause et al., 2000), and diabetes (Hu, 2003). A number of ergonomic interventions have been tested and implemented in an attempt to mitigate these negative consequences, including highly adjustable or alternative office chairs to change sitting postures (Antle et al., 2015; De Carvalho et al., 2016; Le and Marras, 2016; Noguchi et al., 2019), administrative controls such as intermittent movement breaks (Waongenngarm et al., 2021), walking (De Carvalho, 2015; Kar and Hedge, 2020), and alternating between sitting and standing (Gallagher et al., 2014; Karakolis and Callaghan, 2014; McKinnon et al., 2021). However, these interventions are yet to

yield consistent reductions in LBP development in the workplace.

It is widely reported that inconsistent use is a common barrier to adopting workstations that may encourage movement during sedentary work (e.g., sit-stand workstations) (Riddell and Callaghan, 2021; Wilks et al., 2006). The use of sit-stand workstations in a controlled environment shows positive outcomes, including decreased discomfort, decreased sitting time, and increased variation in postures (Gallagher et al., 2014; Husemann et al., 2009; Karakolis and Callaghan, 2014; McKinnon et al., 2021; Robertson et al., 2013). However, a survey conducted for four Swedish companies found that installing sit-stand workstations without training was insufficient to meaningfully alter musculoskeletal loading, with only approximately 20% of workers regularly using the sit-stand workstation for postural transitions daily (Wilks et al., 2006). Implementation of such workstations requires education and/or training. Training has been associated with decreases in sedentary time (Gilson et al., 2012; Graves et al., 2015; Maike Neuhaus et al., 2014a,b), decreases in musculoskeletal discomfort, and increases in utilization (Nerhood and Thompson, 2016; Maike Neuhaus et al., 2014a,b; Wilks et al., 2006). However, these intervention study results continue to show difficulties in maintaining utilization over time and behavioural change that includes frequent movement at the workstation (Riddell and Callaghan, 2021; Wilks et al., 2006). A recent qualitative

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<https://doi.org/10.1016/j.apergo.2023.104044>

Received 23 November 2022; Received in revised form 27 April 2023; Accepted 5 May 2023

Available online 13 May 2023

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evaluation study highlighted two main reasons for non-use: 1) habitual reasons and 2) practical workplace issues (Renaud et al., 2020). The non-users in the study (Renaud et al., 2020) had a persistent habit of sitting that led them to forget to use the standing option, and they did not express a motivation to change or believe in the benefit of standing. The practical issues included the inability to raise the desk because of poor cable management and the time it takes to find the right sitting and standing heights (Renaud et al., 2020).

A novel office chair (Movably Pro, Movably Inc., Boston, USA) was designed to address these reasons for non-use by providing prompts and actively controlled seating changes to facilitate switching postures and eliminating desk/work surface movement when transitioning. The novel chair has mechanical height and footrest adjustments and is positioned higher than a traditional office chair. Because of this higher seating position, it requires a height adjustable or higher work surface. The seat pan is split into the left and right sides, and each side is motorized to move and follow the movement of the user's legs as they transition between sitting and standing. The seat pan is electro-mechanically controlled with manual input (push of a button on the side of the chair) or automatically with a pre-programmed mode (used for the experimental protocol in this study). When using this chair, the desk height is always set at the standing work position, eliminating the need to move the desk height and making transitions easier and faster. This workstation configuration should allow more frequent movement during prolonged sedentary work without affecting task performance. In addition, the novel chair has a backrest that users can use while standing, which could enable users to adopt different supported standing postures. Both features – frequent movement and posture change while standing – have been shown to benefit individuals known to develop LBP during prolonged standing, known as pain developers (PDs) (Gallagher and Callaghan, 2016; McKinnon et al., 2021; Nelson-Wong and Callaghan, 2010a). This novel chair is yet to be evaluated as an office workspace solution with a particular interest in addressing issues with PDs.

The objective of this study was to compare the novel active chair with traditional sitting and standing workstations during a 2-h bout of sedentary desk work. Specifically, the following variables were compared between groups of pain status (PDs and NPDs) and sex (males and females): 1) pelvis and lumbar postures, 2) perceived low back discomfort, and 3) work productivity. It was hypothesized that the novel chair sitting would be comparable to the flexed lumbar posture typically observed in traditional sitting, while the supported standing in the novel chair would result in increased lumbar flexion compared to the more extended posture typically observed in traditional standing. In terms of discomfort, it was hypothesized that the use of the novel chair would result in reduced low back discomfort due to the posture changes that result from frequent movement. The workstation type was not expected to affect task performance.

2. Methods

Three laboratory sessions, each testing a different workstation configuration (active chair, standing, and sitting), were completed by each participant. Participants were instrumented with accelerometers to measure head, spine, and pelvis postures and rigid body clusters to measure trunk and lower extremity postures. During the 2-h collection, participants completed three controlled computer tasks (typing, data entry, and reading comprehension) to simulate desk work.

2.1. Participants

Sixteen participants, eight females (age = 23.5 ± 3.6 years; height = 1.63 ± 0.08 m; mass = 60.8 ± 7.4 kg; BMI = 22.8 ± 2.3 kg/m²) and eight males (age = 25.1 ± 2.4 years; height = 1.74 ± 0.06 m; mass = 76.9 ± 7.8 kg; BMI = 25.3 ± 2.6 kg/m²) were recruited from the university student population for the study via email announcements and

in-class recruitment. The Office of Research Ethics approved all experimental procedures, and participants read and signed the informed consent document before the first laboratory session (#43643). All participants self-reported that they were free of LBP or injuries at the time of participation and the previous six months. Half of the participants were classified as standing-induced transient low back pain developers (PDs) and the other half as non-pain developers (NPDs). The sample size was determined a-priori based on the means and a standard deviation of Visual Analogue Scale (VAS) scores for PDs and NPDs following a 2-h standing protocol, similar to this study (Nelson-Wong and Callaghan, 2010a). The sample size calculation showed that a minimum of seven participants should be included in each group. There was no difference in demographic characteristics (age, height, mass, BMI, baseline low back discomfort before the standing condition) between PD and NPDs ($p > 0.05$) (Table 1).

2.2. Instrumentation

Three-dimensional kinematics of the head, trunk, lumbar spine, pelvis, thighs, and lower legs were measured using an optoelectronic motion capture system (Optotrak Certus, Northern Digital Inc., Waterloo, ON, Canada) and tri-axial accelerometers (ADXL335, Analog Devices, Norwood, MA, USA). Rigid clusters with infrared (iRED) markers were placed on the chest, S1, and bilaterally on the lateral aspects of the thighs and shanks. Following the placement of clusters, a digitizing probe was used to define the endpoints of each body segment. Following digitization, participants completed functional joint trials to identify knee and hip joint centers (Camomilla et al., 2006; Schwartz and Rozumalski, 2005). The marker data were collected at 25 Hz using a 16-bit A/D system (Optotrak Data Acquisition Unit II, Northern Digital Inc., Waterloo, ON, Canada). Accelerometers were placed on the back of the head, C7, L1, and S1. Each accelerometer was used as an inclinometer to measure the cervical (angle between head and C7), thoracic (angle between C7 and L1), lumbar (angle between L1 and pelvis), and pelvis angles. The accelerometer data were collected at 250 Hz using a 16-bit A/D system (Optotrak Data Acquisition Unit II, Northern Digital Inc., Waterloo, ON, Canada). Quiet standing and maximum lumbar flexion trials were collected to normalize postures. The 100 mm VAS was presented digitally on a sliding scale. At each time point, participants were instructed to rate their current level of pain or discomfort in each body region from 0 mm (no pain) to 100 mm (worst pain imaginable).

2.3. Testing protocol

Each participant completed a 3-h laboratory session on three separate days, with each day testing a different workstation configuration. A minimum of 24 h was given between each session. Three randomized workstation conditions were tested: the novel chair sit-stand (Movably Pro, Movably Inc.), traditional standing, and traditional sitting (Embodiment, Herman Miller Inc.) (Fig. 1). In each condition, the office workstation was set up according to the CSA ergonomic guidelines (CSA-Z412, 2000). During workstation setup, participants were also exposed to three standardized office tasks: typing (Mavis Beacon typing training software), data entry (transfer information from invoice to empty form), and reading comprehension (practice Graduate Record Exam). Once the

Table 1
Equivalency of baseline characteristics between NPD and PD.

	NPD (n = 8)	PD (n = 8)	p-value
	Mean (SD)	Mean (SD)	
Age (years)	25.38 (3.07)	23.25 (2.87)	0.12
Height (m)	1.68 (0.11)	1.70 (0.06)	0.87
Mass (kg)	71.18 (13.06)	66.56 (9.02)	0.40
BMI (kg/m ²)	25.10 (2.70)	22.97 (2.45)	0.16
Baseline low back discomfort (mm)	0.84 (2.37)	0.71 (1.17)	0.41

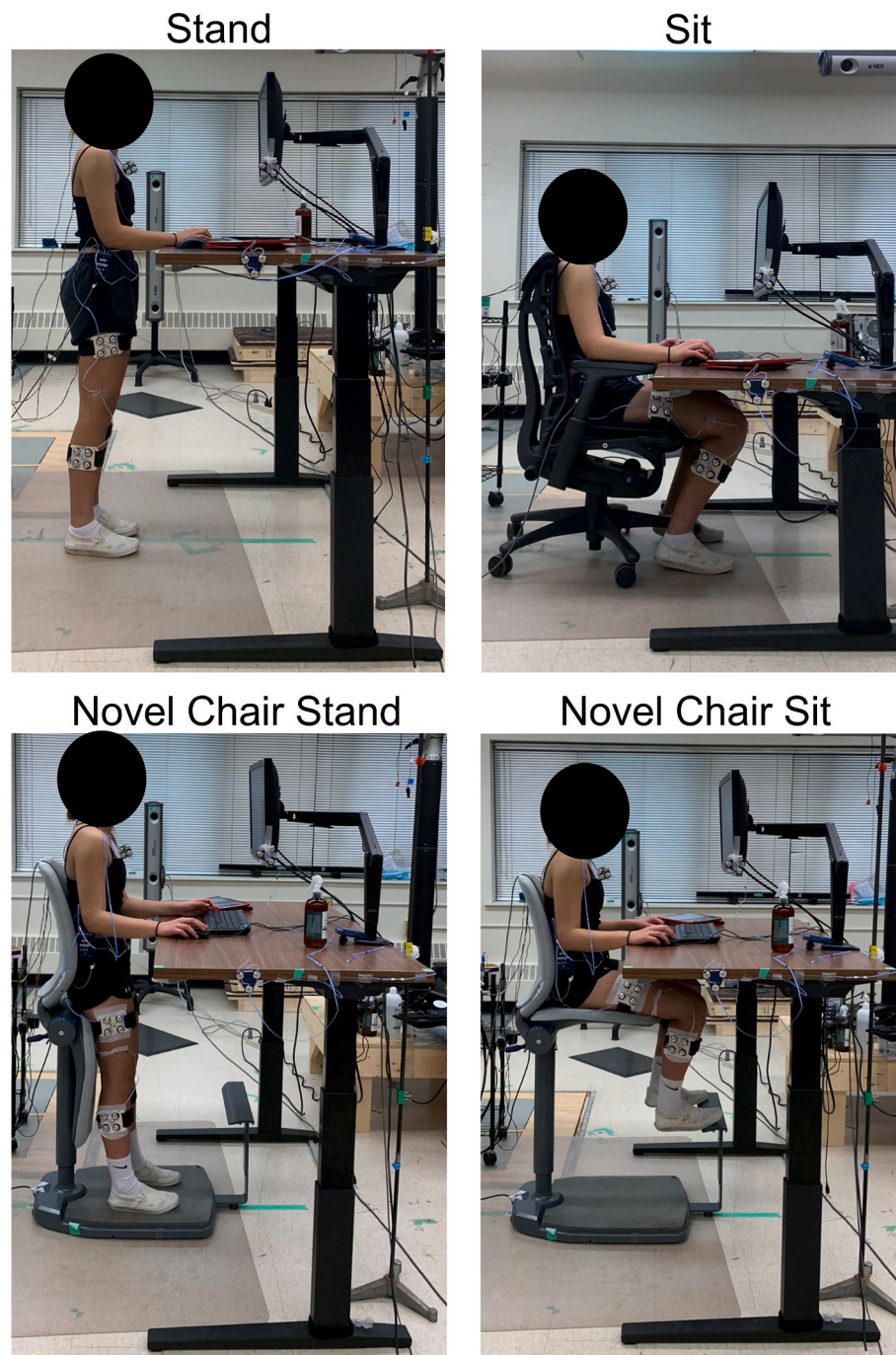


Fig. 1. Workstation conditions tested in this study.

workstation was set up, participants were instrumented for kinematic motion tracking, followed by the 2-h office task protocol. For 2 h, each of the three office tasks was completed twice in a randomized order (six blocks in total). Each block was 15 min in duration, which was followed by a 5-min washout period before the next task started. Both workstation configurations and tasks were block randomized. Discomfort, measured by a VAS, was taken at the baseline (0 min), after each block of office work tasks (at 15, 35, 55, 75, and 95 min), and at the end of the session (115 min). Discomfort development was tracked for 13 body regions: neck, shoulders, left and right upper and lower back, buttocks, upper legs, and lower legs.

Prior to starting the collection with the novel chair, participants were trained to be able to complete smooth transitions between sitting and

standing. Generally, 10 min of training was enough to minimize affordance (e.g., hip hiking and lateral leaning) while maintaining balance to complete both transitions. The novel chair collection started with sitting. The posture alternated between sitting and standing every 3 min, with auditory and proprioceptive cues from the chair, and all transitions began with the movement of the right leg.

2.4. Data analysis

The accelerometer data were processed using a custom Matlab program (The Mathworks Inc., Natick, MA). The data were padded and then smoothed using a dual-pass second-order low-pass Butterworth filter with an effective cutoff frequency of 1 Hz (Howarth and Callaghan,

2009; Smith, 1989; Williams et al., 2022) and calibrated using ± 1 g acceleration. From the calibrated signals, voltage data were converted to angular measures (degrees) using standard four-quadrant trigonometric equations. All flexion-extension angles were represented as angular deviations from an upright standing posture. The lumbar spine angles were represented as the percentage of the maximum flexion angle.

The marker data were also processed using a custom Matlab program (The Mathworks Inc., Natick, MA). The data were padded and then smoothed using a dual-pass fourth-order low-pass Butterworth filter with an effective 1 Hz cutoff frequency. The upright standing calibration trial was used to create segment-fixed local coordinate systems for the trunk, pelvis, thighs, and shanks using the respective anatomical segment endpoints. Given the obstructed line-of-sight to the pelvis cluster during the sedentary exposure (i.e., due to the participant's close interface with the seatback and seat pan), a three-dimensional accelerometer-based pelvis segment was developed and determined based on angular deviations measured with the S1 accelerometer and a three-dimensional vector rotation approach (Zehr et al., 2021). Euler angles of the trunk and thighs (with respect to the global coordinate system) and hips and knees (between respective segments) were calculated with all angles representing angular deviations from the upright standing during the calibration trial.

Normalized kinematic data were used to calculate the averages in each block and the amplitude probability distribution function of the entire sedentary exposure (APDF). It was determined *a priori* that the 10, 50, and 90th percentiles of the APDFs would be analyzed as these are standard measurement points spanning the spectrum of exposure, and they would be separately evaluated since differences between each level were expected. Specifically for the novel chair dataset, two of the 3-min sitting and standing periods in each block (6 min of sitting and 6 min of standing of the 15-min block total) were used to calculate kinematic outputs (i.e., novel chair sit and novel chair stand).

Productivity was assessed for each task block within a workstation configuration session. Typing productivity was evaluated by determining the accuracy (the percentage of words correctly typed) and adjusted words per minute, which is effectively a combined measure of work volume and accuracy. Similarly, the productivity of the data entry task was evaluated based on the accuracy (percentage of correct entries) and adjusted data entry score (the volume of entries adjusted for entry errors). Lastly, reading comprehension was evaluated by the percentage of correct answers.

The discomfort rating was used 1) to distinguish the pain groups and 2) to assess discomfort development over time. The baseline values (0 min) was removed from all subsequent ratings. PDs were categorized *a posteriori* based on their low back discomfort reported during traditional standing, which was one of the workstation conditions in this study. Similar to previous studies, an increase in 10 mm of discomfort level from baseline on a 100 mm visual analog scale (VAS) was used as a threshold for separating participants (Gallagher et al., 2014; Gallagher and Callaghan, 2016; Marshall et al., 2011; Nelson-Wong and Callaghan, 2010a). This threshold is based on the minimally clinically important change (MCIC) outlined by Hägg et al. (2003) and Kelly (1998). For individuals to perceive that the condition has worsened, a change of 8–9 mm on VAS is required (Hägg et al., 2003; Kelly, 1998). This value is greater than the MCIC derived from the standard error of measurement (SEM), which was 2.90 mm. The SEM was calculated using the low back discomfort data from NPDs (no change observed due to standing) and estimating the smallest change possible to detect, with a 95% probability beyond the measurement error (Ostelo and de Vet, 2005). The classification based on a 10 mm threshold has 83% between-day repeatability, suggesting that changes in the results are due to the workstation conditions and not due to variability between test sessions (Nelson-Wong and Callaghan, 2010b).

2.5. Statistical analysis

All statistical analyses were performed using R (R Core Team, 2019) and the RStudio software (Version 1.1.463, RStudio Inc., Boston, MA). The parametric model assumptions were tested using Levene's test, followed by Mauchly's sphericity test. When the assumption of sphericity was violated, corrected p-values from the Greenhouse-Geisser correction were used. For motion data averages and discomfort ratings, four-way mixed ANOVAs with within-factors of Time (6 levels) and Workstation (4 levels; traditional sit, traditional stand, novel chair sit, and novel chair stand) and between-factors of Sex (2 levels) and Pain Status (2 levels) were used for the main analysis ($p < 0.05$). To evaluate the effect of tasks on motion data, all subjects were collapsed into one group, and two-way mixed ANOVAs with within-factor of Workstation (3 levels; traditional sit, traditional stand, and novel chair) and Task (3 levels) were performed ($p < 0.05$). For APDF data and productivity measures, three-way mixed ANOVAs with within-factor of Workstation (3 levels; traditional sit, traditional stand, and novel chair) and between-factors of Sex and Pain Status were used ($p < 0.05$). Two types of post-hoc simple effect tests were performed depending on the outcome of the main analysis: Dunnett's test and pairwise comparisons with Bonferroni corrections.

3. Results

3.1. Pain group classification

Based on the 10 mm low back discomfort threshold, participants were separated into NPD and PD groups. There were eight participants (four males and four females) in each group. No difference was found in age, height, mass, BMI, and baseline low back discomfort between NPD and PD ($p > 0.05$; Table 1). A *t*-test revealed that PDs had a significantly higher low back discomfort score than NPDs after 2 h of standing (12.34 ± 7.93 vs. 0.60 ± 1.44 mm; $p < 0.001$).

3.2. Differences in postures between the novel chair and traditional workstations

Interaction effects containing Workstation were found for the average pelvis tilt angle and normalized lumbar angle (Fig. 2). There was an interaction effect of Workstation x Time for the pelvis tilt angle ($p < 0.001$), where the angle fluctuated over time for the novel chair sitting only. At 75 min, the novel chair standing posture was different from traditional standing and both traditional and novel chair sitting postures ($p < 0.05$). For the lumbar angle, there was an interaction effect of Sex x Pain x Workstation x Time ($p = 0.031$). This interaction was driven mainly by Workstation x Time, similar to pelvis tilt angles. There was a significant change toward less flexion in the normalized lumbar angle from 15 min to 75 min ($p = 0.034$). At this point, both the novel chair sitting and standing postures were different from traditional standing and sitting postures ($p < 0.01$). Both the pelvis tilt and lumbar angles returned to the initial values by the end of the sedentary exposure.

When the data for the sedentary exposures were collapsed over time to assess the distribution using APDF, all three probability magnitudes showed a significant effect of Workstation for both pelvis tilt and lumbar angles ($p < 0.001$) (Fig. 3). For pelvis tilt, the novel chair standing was not different from traditional sitting or traditional standing at the 10th percentile but was different from the novel chair sitting at 10, 50, and 90th percentiles and traditional standing at the 50th and 90th percentiles ($p < 0.01$). The pelvis was more anteriorly tilted in traditional standing compared to the novel chair standing and sitting and traditional sitting ($p < 0.001$). At the 50th and 90th percentiles, the pelvis was more posteriorly tilted in the novel chair standing compared to traditional standing ($p < 0.01$). For the lumbar angle, standing differed from sitting regardless of Workstation ($p < 0.05$). At each level of

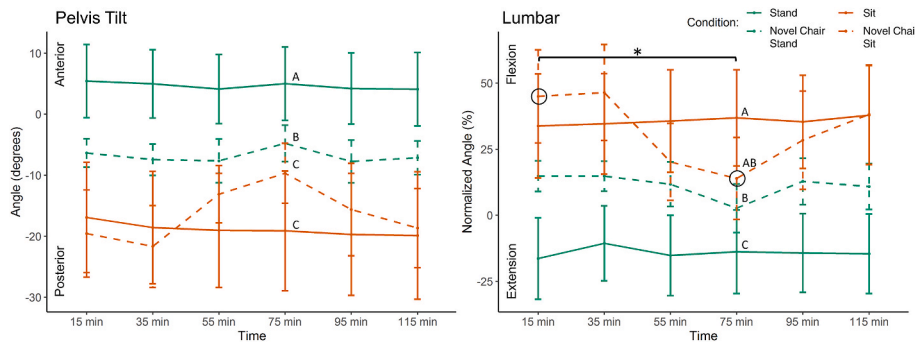


Fig. 2. Average pelvis tilt (left) and lumbar (right) angles over time between four workstation conditions. Standing conditions are in green, and sitting conditions are in orange. Solid lines represent the traditional conditions, and dotted lines represent the novel chair. Significant post-hoc analysis over time (compared to 15 min) within a workstation condition is denoted with asterisks ($p < 0.05$). Significant post-hoc analysis between workstation conditions (within a time frame) is denoted with letters ($p < 0.05$). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

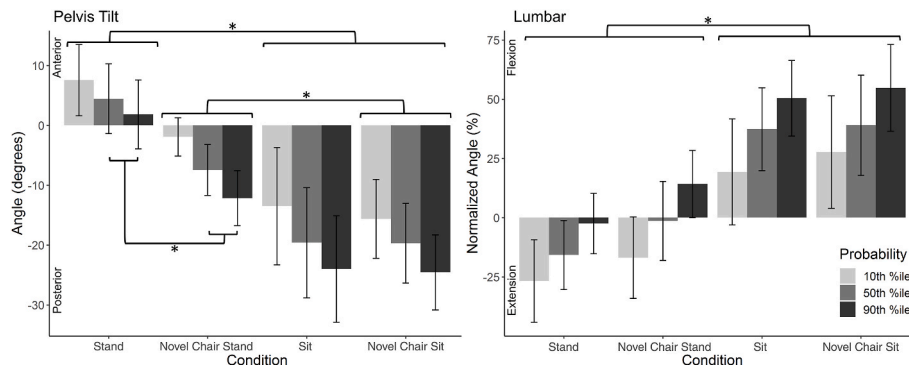


Fig. 3. Pelvis tilt (left) and lumbar (right) angles at the 10th, 50th, and 90th percentile between four workstation conditions. Significant post-hoc analysis between workstation conditions is denoted with asterisks ($p < 0.05$).

probability magnitude, the novel chair sitting resulted in greater posterior pelvic tilt and lumbar flexion compared to traditional sitting; however, this difference ($< 2.5^\circ$ pelvic tilt and $< 8.5\%$ lumbar normalized angle) was not significant. For the thoracic angle, there was no significant result containing the effect of Workstation. Lastly, there was no significant interaction or main effect for pelvis obliquity (lateral flexion) and cervical angles ($p > 0.05$).

3.3. Discomfort development

Only low back and lower leg showed significance containing the effect of Workstation. There were significant interaction effects of Pain Status x Workstation ($p = 0.001$) and Pain Status x Time ($p = 0.038$) for low back discomfort development (Fig. 4). It should be noted that the interaction between pain status and time was expected since the selection of the pain group was based on the discomfort score. Post-hoc analyses comparing workstation conditions within each pain group

showed that there was significantly less discomfort with the novel chair and traditional sitting in PDs compared to traditional standing ($p < 0.01$).

For lower legs, there were significant interaction effects of Workstation x Time ($p = 0.034$) and Sex x Pain x Time ($p = 0.010$) (Fig. 5). Post-hoc analyses comparing workstation conditions at 15 and 115 min showed that at the start, there was no difference between workstation conditions, but by the end of the sedentary exposure, traditional standing had significantly higher discomfort compared to the novel chair and traditional sitting ($p < 0.01$) (Fig. 5). Another region of the body that showed the significance of Time was the buttocks ($p = 0.008$), where the increase in discomfort became significant at 115 min ($p = 0.049$), with the novel chair having the least change in discomfort over the 2 h (Fig. 5).

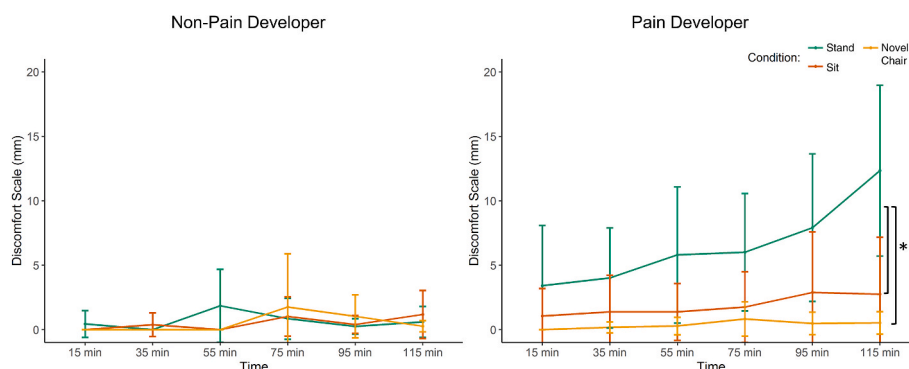


Fig. 4. Low back discomfort score over time for NPDs (left) and PDs (right). Each line represents different workstation conditions. Significant post-hoc analysis between workstation conditions is denoted with asterisks ($p < 0.05$).

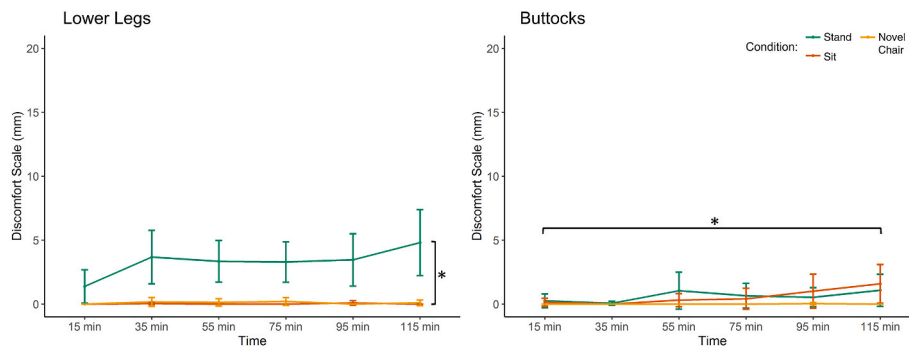


Fig. 5. Lower leg (left) and buttocks (right) discomfort score over time. Significant post-hoc analysis between workstation conditions (for lower legs) and over time (for buttocks) is denoted with asterisks ($p < 0.05$).

3.4. Task performance

There was no significant effect of Workstation on task-specific kinematics or task performance, except for the adjusted data entry score (Fig. 6). There was an interaction effect of Sex x Pain Status x Workstation ($p = 0.031$), in which the female PDs scored significantly less than the female NPDs in traditional standing ($p = 0.036$).

4. Discussion

As hypothesized, the novel chair facilitated an intermediate posture and reduced low back discomfort without compromising task performance among the young, healthy population during a 2-h sedentary exposure. While novel chair sitting was not different from traditional sitting, novel chair standing produced an intermediate posture in which the pelvis and lumbar angles fell between traditional sitting and standing. Low back discomfort in the novel chair remained below the threshold of 10 mm in 100% of PDs, which has not been observed nor reported in previous intervention studies on a PD cohort (Gallagher and Callaghan, 2016; McKinnon et al., 2021; Nelson-Wong and Callaghan, 2010a). Task performance was unaffected by the use of the novel chair, even though each participant performed 30 transitions during six 15-min blocks of office tasks.

All participants who were classified as PDs reported less than 10 mm of low back discomfort in the novel chair, suggesting that this approach may be a viable solution to minimizing the negative effects of sedentary work. To the best of the author's knowledge, this is the first study to demonstrate that 100% of PDs in traditional standing can be classified as NPDs with an intervention in a controlled environment (i.e., all PDs responded with less than 10 mm of low back discomfort throughout the

2-h sedentary exposure in the novel chair). Gallagher and Callaghan (2016) demonstrated a reduction of low back discomfort from 16.2 mm to 6.8 mm on average with a standing aid; however, when examined individually, 33% of participants remained as PDs with this intervention. Nelson-Wong and Callaghan (2010a) also used a standing aid to successfully reduce low back discomfort from 22.8 mm to 9.25 mm, but it remains unclear if this response was consistent across all PDs. McKinnon et al. (2021) showed that early frequent movement led to reduced low back discomfort within the first hour of standing (6 mm for PDs), but at the end of the 124-min exposure, PDs reported discomfort exceeding the threshold of 10 mm on average. The peak low back discomfort in the novel chair for PDs was 0.83 (SD 1.60 mm), which is notably lower than the aforementioned studies and compared to traditional standing (12.33 (SD 7.93) mm) and sitting (2.88 (SD 5.63) mm). Although PDs did not exceed the threshold of 10 mm with the novel chair (MCIC for worsening condition), the magnitude of reduction in discomfort score from traditional standing to the novel chair also did not meet the 15–20 mm MCIC criteria for clinically important improvement (Hägg et al., 2003; Olsen et al., 2017; Ostelo and de Vet, 2005; Sokka, 2005). It should be noted that MCIC is often assessed using a clinical population experiencing chronic or acute pain. The participants in this study were healthy as evidenced by the similar baseline score of PDs and NPDs (0.71 mm and 0.84 mm, respectively). A systematic review by Olsen et al. (2017) found a strong association between the absolute value of the baseline pain score and the magnitude of reduction needed to perceive meaningful improvement. According to the meta-analysis of individual patient data from seven studies (918 patients), for each 10 mm increase in baseline pain, the MCIC increased by 3.1 mm (Olsen et al., 2017). For individuals with a baseline score of <40 mm (125 patients), the average MCIC was 5.75 mm (Fig. 3 in Olsen et al. (2017)).

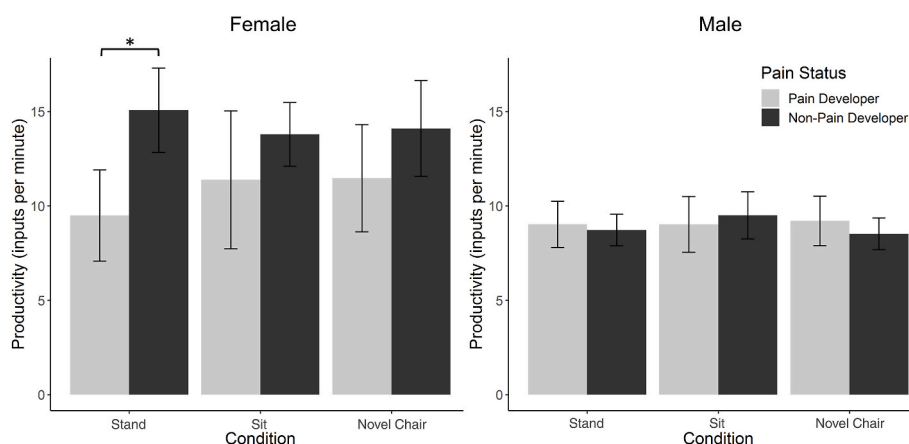


Fig. 6. The adjusted data entry score for females (left) and males (right). PDs are in light grey, and NPDs are in dark grey. Significant post-hoc analysis between pain status groups is denoted with asterisks ($p < 0.05$).

The PDs in this study saw an improvement of 11.90 (SD 5.28) mm with the novel chair compared to the traditional standing, which is sufficiently larger than the MCID of 5.75 mm, suggesting that this improvement in PDs is likely clinically meaningful and relevant. Throughout the duration of the sedentary work exposure, the novel chair demonstrated the lowest low back discomfort level, though this level was not statistically different from traditional sitting. The novel chair also had reduced lower leg discomfort compared to traditional standing over time, which may be a reflection of less blood pooling and increased venous return due to frequent movement (Antle et al., 2015). These results from low back and lower leg discomfort suggest that the novel chair is similar to traditional sitting even though individuals moved frequently and stood for 50% of the sedentary exposure. A 50% reduction in sitting is considerable since among successful intervention studies, the average sitting time reduced was 77 min per 8 h work day (Maike Neuhaus et al., 2014a,b), which still equals to approximately 65% (range 45–72%) of work hours spent in sitting (Alkhajah et al., 2012; Ellegast et al., 2012; Healy et al., 2013; Maike Neuhaus et al., 2014a,b; Parry et al., 2013; Pronk et al., 2012).

The novel chair enabled users to adopt postures that are similar to a traditional sit-stand workstation as well as postures in between sitting and standing. There has been an increasing interest in varying tissue loading during sedentary work by adopting postures different from traditional sitting and standing (Antle et al., 2015; Gallagher et al., 2014; Karakolis et al., 2016; Le and Marras, 2016; Noguchi et al., 2019) since prolonged sitting and standing can lead to the development of LBP. A possible cause of sitting LBP is linked to increased lumbar flexion (De Carvalho et al., 2010), which leads to increased tissue creep (Howarth et al., 2013) and migration of nucleus pulposus to the posterior region (Alexander et al., 2007). Possible causes of standing LBP include abnormal neuromuscular control (e.g., gluteus medius co-activation) (Marshall et al., 2011; Nelson-wong et al., 2012; Nelson-Wong et al., 2009, 2008; Nelson-Wong and Callaghan, 2010c) and greater deviation from the neutral zone towards extension (Gallagher, 2014), which may lead to facet capsule irritation. One way to reduce prolonged sitting or standing without taking a break is to alternate these postures using a sit-stand workstation. Using a conventional sit-stand workstation with a

15:5 min sit-to-stand rotation resulted in lower LBP and reduced lumbar flexion in sitting; however, this postural change did not result in LBP lower than 10 mm nor changes in compressive or shear loading (Karakolis et al., 2016). Similarly, LBP was temporarily reduced but not mitigated by 45:15 min stand-to-sit rotation (Gallagher et al., 2014). As with the novel chair, the sitting lumbar posture was not different from traditional sitting (32.01% and 35.68%, respectively). It should be noted that the range in which the lumbar posture varied in the novel chair was greater than that of traditional sitting (32.52% and 4.06%, respectively). Another way to reduce prolonged sitting or standing is to adopt a hybrid posture. It has been hypothesized that sitting with a greater trunk-thigh angle (i.e., perching) would bring the lumbopelvic posture closer to its neutral position (Keegan, 1953); however, as long as the pelvis was in contact with the surface, it tended to rotate posteriorly, resulting in increased lumbar flexion (Noguchi et al., 2019). Even when a significant change in lumbar angle was achieved at $>145^\circ$ trunk-thigh angle, it came at the cost of increased lower limb effort to maintain balance (Noguchi et al., 2019). Interestingly, the standing posture adopted in the novel chair fell in the middle of traditional standing and sitting (lumbar posture: 11.27%, compared to -14.17% and 35.68% in traditional standing and sitting, respectively). The magnitude of change in normalized lumbar angle, 24.41% change from traditional sitting, was comparable to Noguchi et al. (2019) for the intermediate postures linked to significant changes in lumbar spine postures from sitting and therefore considered meaningful. The posture likely comes from participants leaning against the novel chair's backrest resulting in posterior pelvis rotation and a reduction of lumbar extension. It should be noted that all participants chose to use the backrest when standing, without any prompt to do so (Fig. 7). This posture was achievable without increasing the lower limb effort, as evidenced by the low level of lower limb discomfort reported by participants. Therefore, adopting a posture without restricting the movement of the pelvis (i.e., without a seat pan such as the case in novel chair standing) can result in a favorable outcome in terms of kinematics and discomfort.

Worker productivity is often raised as an additional concern when it comes to introducing a new workstation (Karakolis and Callaghan, 2014), but frequent movement and adopting an alternative posture for

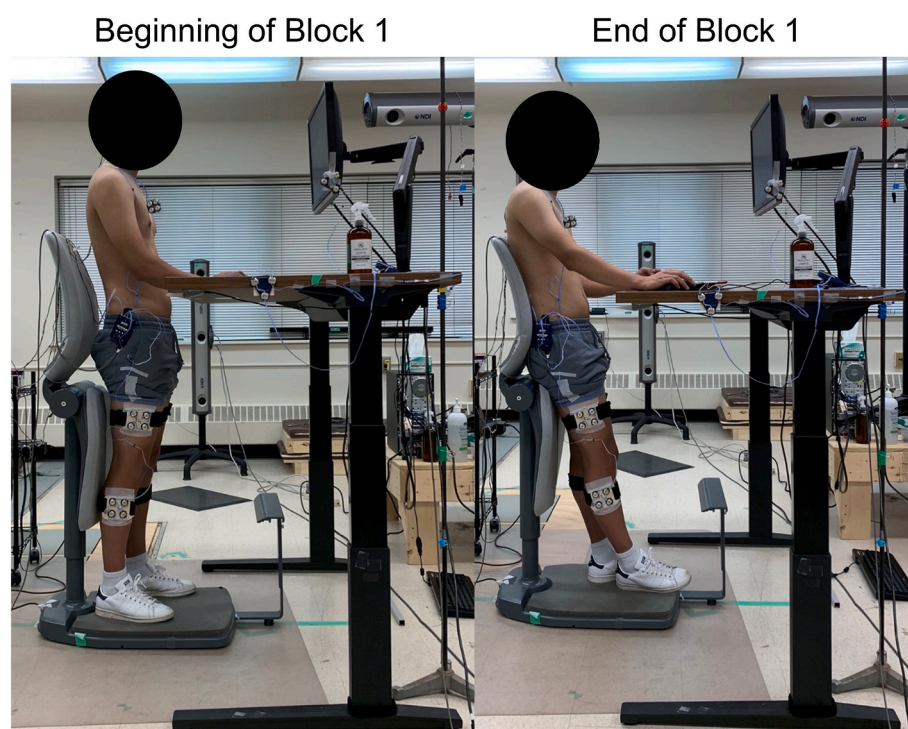


Fig. 7. An example of a participant starting with an upright standing posture and quickly adopting the “lean” standing posture within 15 min.

half of the sedentary exposure in the novel chair did not negatively affect task performance. In this study, participants transitioned approximately 15 times per hour (sitting bout of 3 min), which is more than twice the amount typically reported (3–6 transitions per hour; sitting bout of 15–50 min) in previous studies (Danquah et al., 2017; Huysmans et al., 2019). Despite this increase in the frequency of disruption, participants were able to continue working during transitions, most likely because the workstation height was unchanged (i.e., head location/gaze angle unchanged), and the transitions were prompted and automated. It has been suggested that an increase in discomfort may lead to a decrease in productivity (Haynes and Williams, 2008; Liao and Drury, 2010). In line with previous findings, the only significant interaction effect found in this study was the reduced data entry productivity among female PDs compared to female NPDs in the traditional standing. Productivity loss can also occur as a result of time loss associated with desk height change for sit-stand workstations (Liao and Drury, 2010) or frequent movement. In this study, there was no difference in task performance between workstation conditions, which suggests that although productivity was not improved with a decrease in discomfort, frequent movement did not negatively affect the performance with the novel chair. Given that female PDs performed worse than NPDs, a workspace solution, such as this novel chair, that can achieve a reduction in low back discomfort, can encourage movement, and does not interfere with productivity may be appropriate.

There are several limitations that should be addressed when considering the results of this study. In this study, a direct comparison was not made between a traditional sit-stand workstation and the novel sit-stand chair with a static desk position. Since this level of frequent transition (every 3 min) has never been tested with traditional workstations in previous studies, we cannot eliminate the possibility that frequent movement alone could result in discomfort reduction. However, transitioning at a traditional workstation takes approximately 10–20 s per transition, which equals 5–10% of work time lost if the same protocol (a 3-min interval) were maintained. Therefore, loss in productivity seems likely even if discomfort was eliminated by frequent movement in a traditional workstation. Furthermore, both posture and movement were altered by the novel chair, making it difficult to conclude how each factor contributed independently to changes in discomfort. Both factors have been linked to a reduction in discomfort (De Carvalho, 2015; Gallagher and Callaghan, 2016; Karakolis et al., 2016; Nelson-Wong and Callaghan, 2010a), and therefore, it is plausible that both factors played a role in reducing lower back and lower leg discomfort. The study protocol was developed based on the premise that removing one of the barriers to using the traditional sit-stand desk (i.e., moving the desk to transition) would lead to more frequent transitions. It should be acknowledged that the novel sit-stand chair has not been externally validated to confirm this assumption. Lastly, the results from the young, healthy population exposed to a laboratory-controlled protocol may not extrapolate to the population at large and in a field setting. Different age groups, work tasks, frequent transitions, and longer duration could have different effects on kinematics, discomfort, and task performance. However, compared to the traditional sit-stand workstation, this novel chair requires 56% less work and 64% less energy to transition between sitting and standing (Zehr et al., 2022), suggesting that the novel chair may be energetically favorable if a longer duration of work activity was maintained.

5. Conclusions

Overall, the novel chair facilitated an intermediate posture with frequent transitions between sitting and standing without compromising task performance in a young, healthy population during the 2-h sedentary exposure. The standing posture in the novel chair was an intermediate posture in which the pelvis and lumbar angles fell in between traditional sitting and standing. The sitting posture in the novel chair was comparable to the posture in a traditional chair. There was a

significant reduction in the lower leg and low back discomfort among standing PDs with the novel chair compared to traditional standing, which supports the novel chair as a sound solution for encouraging movement in those who are intolerant to standing. The question remains whether the frequent movement or altered posture through the use of the backrest in standing contributed more to these decreases in discomfort in the novel chair.

Disclosure statement

No conflict of interest was reported by the authors.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: This work was commissioned and partially supported financially by the industrial partner. The study was designed by the investigators and received approval from the industrial partner. The decision to publish any findings, **positive or negative**, was agreed upon **prior** to data collection. The investigators do not own any stock or have financial interest in the company and will not financially or personally gain from this publication.

Acknowledgements

The authors acknowledge funding from the Natural Science and Engineering Research Council of Canada (NSERC). Dr. Jack Callaghan is supported by the Canada Research Chair in Spine Biomechanics and Injury Prevention. This work was commissioned and partially supported financially by the industrial partner. The study was designed by the investigators and received approval from the industrial partner. The decision to publish any findings, positive or negative, was agreed upon prior to data collection. The final manuscript was reviewed and approved by the industrial partner prior to submission.

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